DOM

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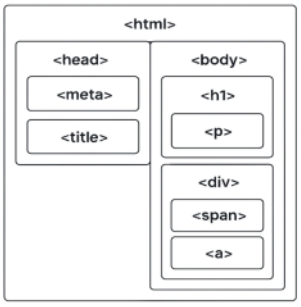
What is DOM?

DOM Stands for **Document Object Model.**

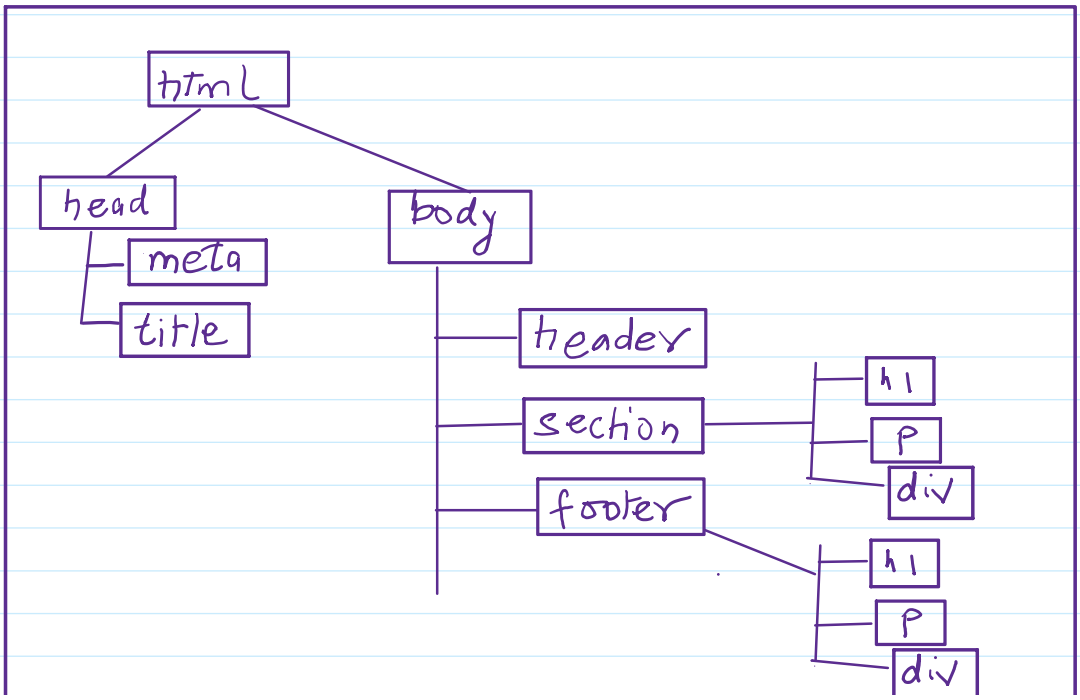
The DOM is a tree-like structure that the browser creates from your HTML page.

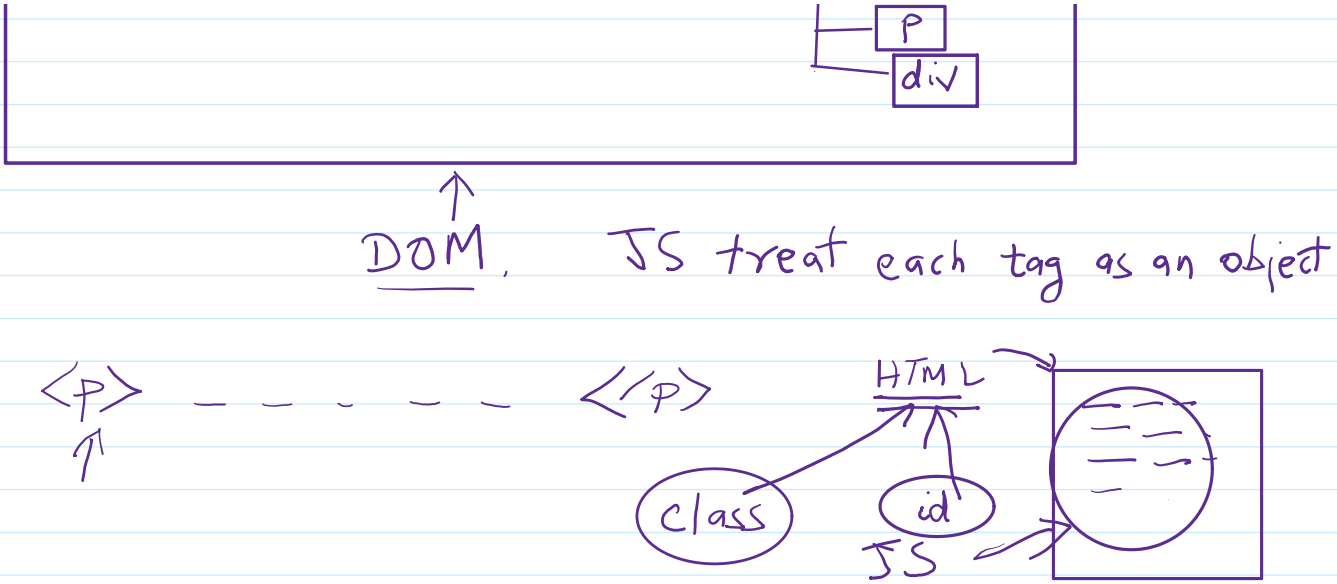
It represents everything on the webpage — text, buttons, images, sections — as objects that JavaScript can access and control.




```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Document</title>
</head>
<body>

</body>
</html>
```





Why We Need DOM?

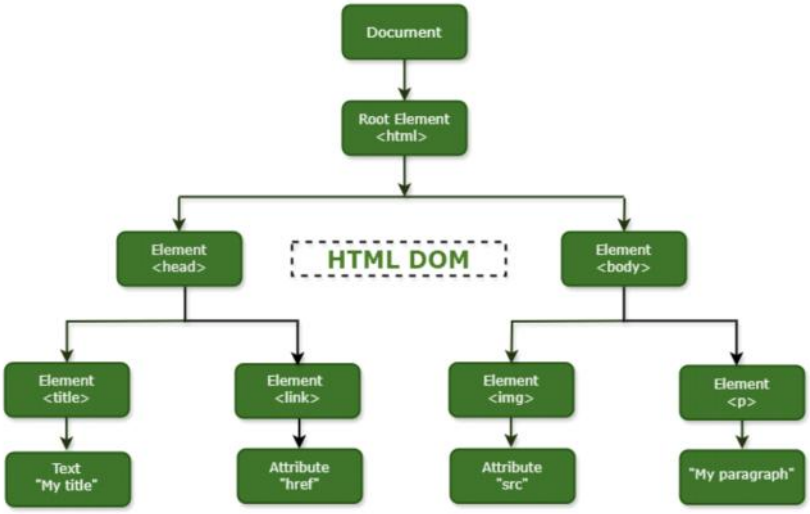
HTML creates the structure of a web page, but **once the page loads**, it's just static.
You can't change content, read input values, or respond to user actions with HTML alone.

Example: You can write a button in HTML, but without the DOM, JavaScript has no way to know when it's clicked or what to do after.

The DOM lets JavaScript:

- Read and change content on the page
- React to user actions (clicks, typing, scrolling)
- Create new elements or remove old ones
- Build dynamic, interactive web pages without reloading

DOM Tree Structure



DOM Tree Structure : Explanation

Root Level – Document Object

- The topmost node is **Document**.
- It represents the entire HTML page loaded in the browser.

<html> – Root Element

- The Document points to the **<html> element**, which is the root of the HTML structure.
- All content on the page is nested inside <html>.

Inside <html> – Two Main Branches

- <head>: Contains metadata and resources.
- <body>: Contains visible content of the page.

DOM Tree Structure

HEAD Section Breakdown

- **<head> → <title>**
  - <title> is a child of <head>.
  - It contains **text**: "My title" (this appears in the browser tab).

- **<head> → <link>**
  - <link> is another child of <head>.
  - It includes an **attribute**: "href" — typically used to link CSS files.

DOM Tree Structure

BODY Section Breakdown

- **<body> → <img>**
  - <img> is a child of <body>.
  - It has an **attribute**: "src" — which defines the image source.

- **<body> → <p>**
  - <p> is also a child of <body>.
  - It contains a **text node**: "My paragraph" — this appears as visible paragraph content.

DOM Tree Structure

BODY Section Breakdown

- **<body> → <img>**
  - <img> is a child of <body>.
  - It has an **attribute**: "src" — which defines the image source.

- **<body> → <p>**
  - <p> is also a child of <body>.
  - It contains a **text node**: "My paragraph" — this appears as visible paragraph content.

Types of DOM Nodes

Node type	Example
Document Node	Entire Webpage
Element Node	<p>
Text Node	Text inside element
Attribute Node	id, class

```
document.getElementById("btn").onclick = function(){
   alert("Button clicked");
}
```

DOM Manipulation = Dynamic Webpages

Without DOM
- Page is static

With DOM

- Content updates live
- Interactive UI
- Realtime response to user actions

Examples:

  - Form validation
  - Mobile menu toggle
  - Dark mode switch
  - Dynamic content loading

Key Properties Used in DOM

Property	Purpose
innerHTML	Get/set HTML element
textContext	Get/set text
style	Modify CSS
classList	Add/remove classes
parentNode	Access parent
children	Access child elements